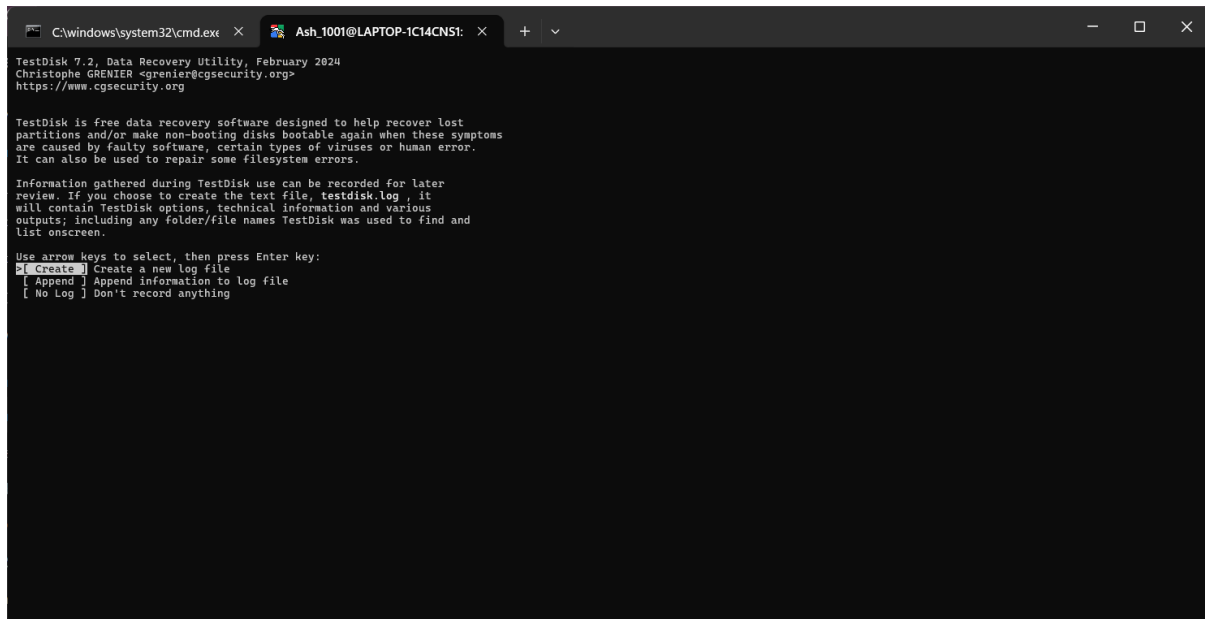


Ex.No:2:- Recover deleted or damaged files from a storage device using Test Disk

Description:

TestDisk step by step to recover a missing partition and repair a corrupted one.

Log creation



```
C:\windows\system32\cmd.exe x Ash_1001@LAPTOP-1C14CNS1: x + v
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

TestDisk is free data recovery software designed to help recover lost
partitions and/or make non-booting disks bootable again when these symptoms
are caused by faulty software, certain types of viruses or human error.
It can also be used to repair some filesystem errors.

Information gathered during TestDisk use can be recorded for later
review. If you choose to create the text file, testdisk.log , it
will contain TestDisk options, technical information and various
outputs; including any folder/file names TestDisk was used to find and
list onscreen.

Use arrow keys to select, then press Enter key:
[ Create ] Create a new log file
[ Append ] Append information to log file
[ No Log ] Don't record anything
```

- Choose Create to instruct Testdisk to create a [log file](#) containing technical information and messages, unless you have a reason to append data to the [log](#) or you execute TestDisk from read only media and must create the [log](#) elsewhere.
- Choose None if you do not want messages and details of the process to be written into a [log file](#) (useful if for example Testdisk was started from a read-only location).
- Press Enter to proceed.

Disk selection

All hard drives should be detected and listed with the correct size by TestDisk:

```
C:\windows\system32\cmd.exe x Ash_1001@LAPTOP-1C14CNS1: x + v
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

TestDisk is free software, and
comes with ABSOLUTELY NO WARRANTY.

Select a media and choose 'Proceed' using arrow keys:
Disk /dev/sda - 374 MB / 356 MiB (RO) - Virtual Disk
Disk /dev/sdb - 167 MB / 159 MiB (RO) - Virtual Disk
Disk /dev/sdc - 2147 MB / 2048 MiB - Virtual Disk
Disk /dev/sdd - 1899 GB / 1824 GiB - Virtual Disk
Disk /dev/loop0 - 536 MB / 512 MiB
Disk /dev/loop01 - 535 MB / 511 MiB

[Proceed] [Quit]

Note:
Disk capacity must be correctly detected for a successful recovery.
If a disk listed above has an incorrect size, check HD jumper settings and BIOS
detection, and install the latest OS patches and disk drivers.
```

- Use up/down arrow keys to select your hard drive with the lost partition/s.
- Press Enter to Proceed.

X If available, use raw device `/dev/rdisk*` instead of `/dev/disk*` for faster data transfer.

Partition table type selection

TestDisk displays the partition table types.

```
C:\windows\system32\cmd.exe x Ash_1001@LAPTOP-1C14CNS1: x + v
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk /dev/loop0 - 536 MB / 512 MiB

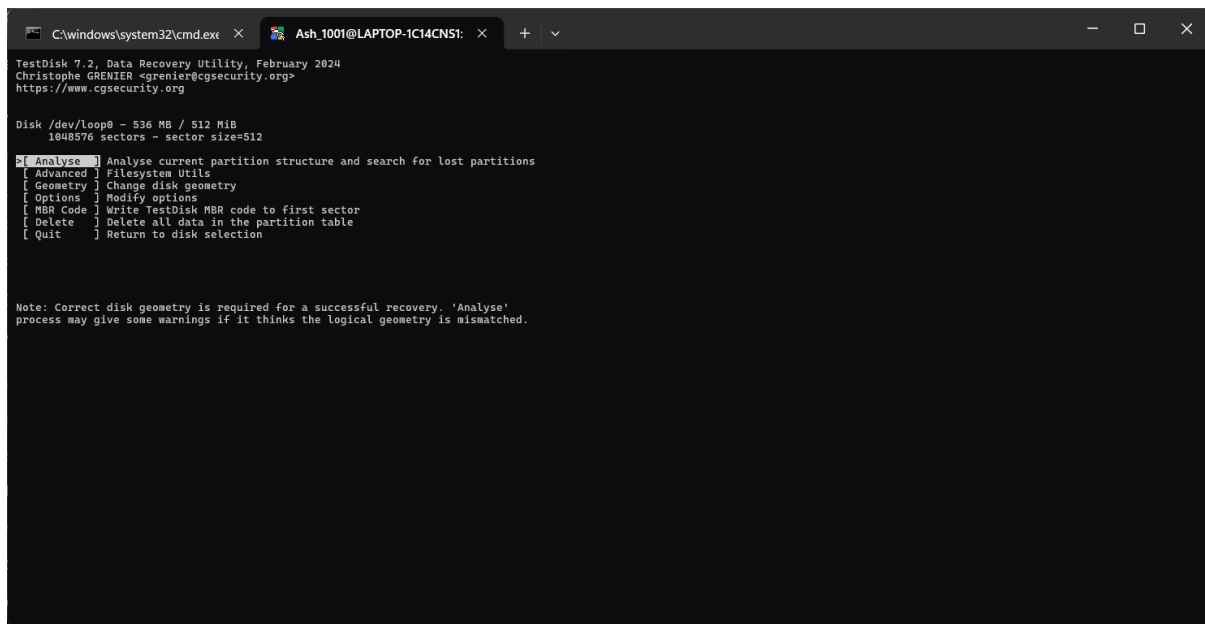
Please select the partition table type, press Enter when done.
[Intel] Intel/PC partition
[EFI GPT] EFI GPT partition map (Mac i386, some x86_64...)
[HuMAX] HuMAX partition table
[Mac] Apple partition map (legacy)
[None] Non partitioned media
[Sun] Sun Solaris partition
[XBox] XBox partition
[Return] Return to disk selection

Hint: Intel partition table type has been detected.
Note: Do NOT select 'None' for media with only a single partition. It's very
rare for a disk to be 'Non-partitioned'.
```

- Select the partition table type - usually the default value is the correct one as TestDisk auto-detects the partition table type.
- Press Enter to Proceed.

Current partition table status

TestDisk displays the menus (also see [TestDisk Menu Items](#)).



```
C:\windows\system32\cmd.exe x Ash_1001@LAPTOP-1C14CNS1: x + v
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

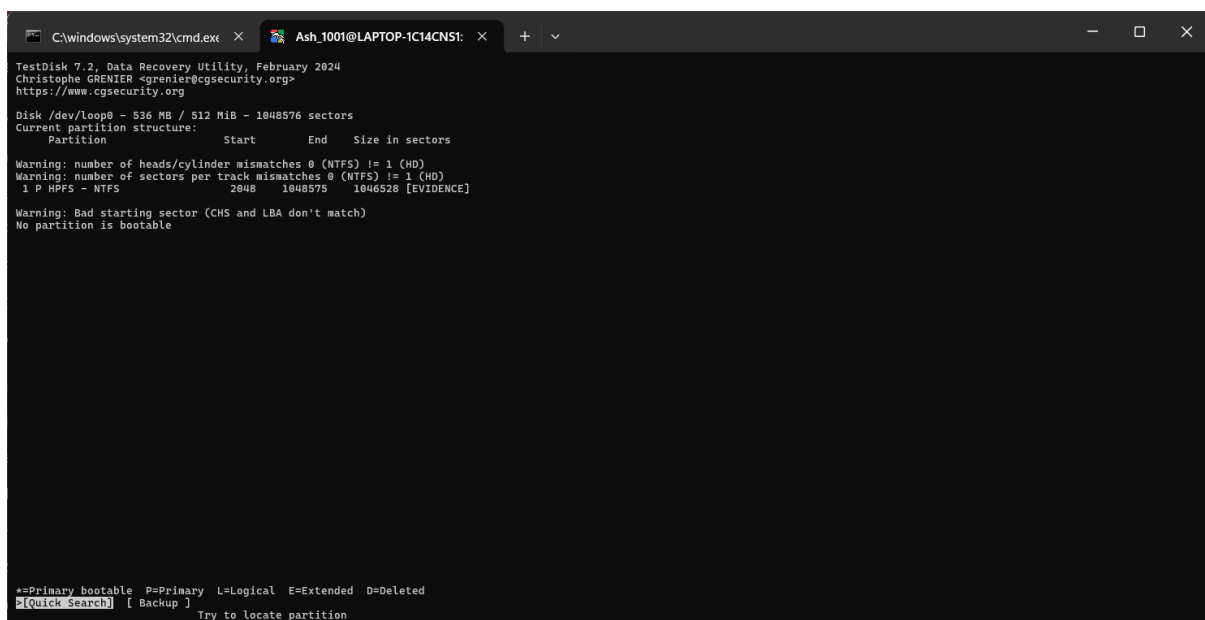
Disk /dev/loop0 - 536 MB / 512 MiB
1048576 sectors - sector size=512

[ Analyze ] Analyze current partition structure and search for lost partitions
[ Advanced ] Filesystem Utils
[ Geometry ] Change disk geometry
[ Options ] Modify options
[ MBR Code ] Write TestDisk MBR code to first sector
[ Delete ] Delete all data in the partition table
[ Quit ] Return to disk selection

Note: Correct disk geometry is required for a successful recovery. 'Analyze'
process may give some warnings if it thinks the logical geometry is mismatched.
```

- Use the default menu "Analyze" to check your current partition structure and search for lost partitions.
- Confirm at Analyze with Enter to proceed.

Now, your current partition structure is listed. Examine your current partition structure for missing partitions and errors.



```
C:\windows\system32\cmd.exe x Ash_1001@LAPTOP-1C14CNS1: x + v
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk /dev/loop0 - 536 MB / 512 MiB - 1048576 sectors
Current partition structure:
  Partition      Start      End      Size in sectors
Warning: number of heads/cylinder mismatches 0 (NTFS) != 1 (HD)
Warning: number of sectors per track mismatches 0 (NTFS) != 1 (HD)
1 P HPFS - NTFS      2048      1048575      1046528 [EVIDENCE]
Warning: Bad starting sector (CHS and LBA don't match)
No partition is bootable

**Primary bootable P=Primary L=Logical E=Extended D=Deleted
[Quick Search] [Backup]
Try to locate partition
```

The first partition is listed twice which points to a corrupted partition or an invalid partition table entry.

Invalid NTFS boot points to a faulty NTFS boot sector, so it's a corrupted filesystem.

Only one logical partition (label Partition 2) is available in the extended partition. One logical partition is missing.

- Confirm at **Quick Search** to proceed.

Quick Search for partitions

TestDisk displays the first results in real time.



(click on thumb to display the image).

During the **Quick Search**, TestDisk has found two partitions including the missing logical partition labeled **Partition 3**.

```
C:\windows\system32\cmd.exe x Ash_1001@LAPTOP-1C14CNS1: x + v
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk /dev/loop0 - 536 MB / 512 MiB - 1048576 sectors
Warning: the current number of heads per cylinder is 1
but the correct value may be 128.
You can use the Geometry menu to change this value.
It's something to try if
- some partitions are not found by TestDisk
- or the partition table can not be written because partitions overlap.

[Continue]
```

- Highlight this partition and press **p** to list your files (to go back to the previous display, press q to Quit, Files listed in red are deleted entries).

All directories and data are correctly listed.

- Press Enter to proceed.

Save the partition table or search for more partitions?

```
C:\windows\system32\cmd.exe x Ash_1001@LAPTOP-1C14CNS1: x + v
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk /dev/loop0 - 536 MB / 512 MiB - 1048576 sectors
Partition      Start      End      Size in sectors
-----
1  0  1048576  1048576  512 MiB

Structure: Ok. Use Up/Down Arrow keys to select partition.
Use Left/Right Arrow keys to change partition characteristics.
+Primary bootable #Primary L=Logical E=Extended D=Deleted
Keys A: add partition, L: load backup, T: change type, P: list files,
Enter: to continue
NTFS, blocksize=4096, 536 MB / 512 MiB
```

- **When all partitions are available** and data correctly listed, you should go to the menu **Write** to save the partition structure. The menu Extd Part gives you the opportunity to decide if the extended partition will use all available disk space or only the required (minimal) space.
- **Since a partition, the first one, is still missing**, highlight the menu **Deeper Search** (if not done automatically already) and press Enter to proceed.

A partition is still missing: Deeper Search

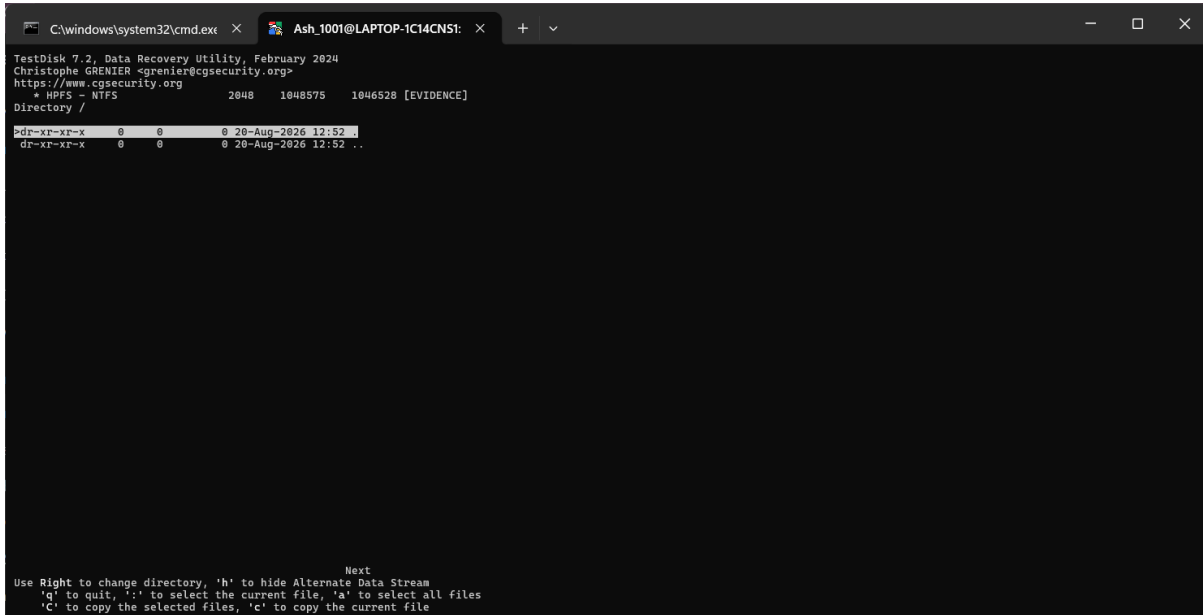
Deeper Search will also search for FAT32 backup boot sector, NTFS backup boot superblock, ext2/ext3 backup superblock to detect more partitions,

it will scan each cylinder  (click on thumb).

After the Deeper Search, the results are displayed as follows:

The first partition "**Partition 1**" was found by using backup boot sector. In the last line of your display, you can read the message "**NTFS found using backup sector!**" and the size of your partition. The "partition 2" is displayed twice with different size.

Partitions listed as D(eleted) will not be recovered if you let them listed as deleted. Both partitions are listed with status **D** for deleted, because they overlap each other. You need to identify which partition to recover.



```

TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
* HPFS - NTFS          2048      1046528 [EVIDENCE]
Directory /
dr-xr-xr-x  0  0      0 20-Aug-2026 12:52
dr-xr-xr-x  0  0      0 20-Aug-2026 12:52 ..

Next
Use Right to change directory, 'h' to hide Alternate Data Stream
'q' to quit, ':' to select the current file, 'a' to select all files
'C' to copy the selected files, 'c' to copy the current file

```

- Highlight the first partition Partition 2 and press **p** to list its data.

The file system of the upper logical partition (label Partition 2) is damaged

 (click on thumb).

- Press **q** for Quit to go back to the previous display.

- Let this partition Partition 2 with a damaged file system marked as D(deleted).
- Highlight the second partition Partition 2 below
- Press p to list its files.

```

TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk /dev/loop0 - 536 MB / 512 MiB - 1048576 sectors

Partition                Start      End      Size in sectors
-----                -
> 1 P HPFS - NTFS         2048      1048575  1046528 [EVIDENCE]

[ Type ] > [ Boot ] [ List ] [Undelete] [Image Creation] [ Quit ]
Boot sector recovery

```

It works, your files are listed, you have found the correct partition!

- Use the left/right arrow to navigate into your folders and watch your files for more verification

Note: FAT directory listing is limited to 10 clusters - some files may not appear but it doesn't affect recovery.

- Press q for Quit to go back to the previous display.
- The available status are Primary, * bootable, Logical and Deleted.

Using the left/right arrow keys, change the status of the selected partition from D(eleted) to **L(ogical)**. This way you will be able to recover this partition.

Hint: read [How to recognize primary and logical partitions?](#)

Note: If a partition is listed *(bootable) but if you don't boot from this partition, you can change it to **P**Primary partition.

- Press Enter to proceed.

Partition table recovery

It's now possible to write the new partition structure.

Note: The extended partition is automatically set. TestDisk recognizes this using the different partition structure.

```
C:\windows\system32\cmd.exe x Ash_1001@LAPTOP-1C14CNS1: x + v
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
1 P HPFS - NTFS 2048 1048575 1046528 [EVIDENCE]
Deleted files
./deleted_evidence.txt 20-Aug-2026 13:07 42

Use : to select the current file, a to select/deselect all files,
C to copy the selected files, c to copy the current file, q to quit
```

- If **all partitions are listed** and only in this case, confirm at **Write** with Enter, y and OK.

Now, the partitions are registered in the partition table.

NTFS Boot sector recovery

The boot sector of the first partition named Partition 1 is still damaged. It's time to fix it. The status of the NTFS boot sector is bad and the backup boot sector is valid. Boot sectors are not identical.

```
C:\windows\system32\cmd.exe x Ash_1001@LAPTOP-1C14CNS1: x Ash_1001@LAPTOP-1C14CNS1: x + v
TestDisk 7.2, Data Recovery Utility, February 2024
Please select a destination where the marked files will be copied.
Keys: Arrow Keys to select another directory
C when the destination is correct
Q to quit
Directory /
dfwxixt-x 0 0 4096 20-Aug-2026 12:50
dfwxixt-x 0 0 4096 20-Aug-2026 12:50
dfwxixt-x 0 0 4096 16-Jun-2026 15:44 boot
dfwxixt-x 0 0 3708 20-Aug-2026 13:06 dev
dfwxixt-x 0 0 4096 20-Aug-2026 13:12 etc
dfwxixt-x 0 0 4096 15-Jul-2026 22:12 home
dfwx----- 0 0 16384 15-Jul-2026 21:56 lost+found
dfwxixt-x 0 0 4096 2-Jul-2026 18:16 media
dfwxixt-x 0 0 4096 28-Jul-2026 09:37 mnt
dfwxixt-x 0 0 4096 2-Jul-2026 18:16 opt
df-xixt-x 0 0 0 20-Aug-2026 12:38 proc
dfwx----- 0 0 4096 2-Jul-2026 18:17 root
dfwxixt-x 0 0 480 20-Aug-2026 12:53 run
dfwxixt-x 0 0 4096 2-Jul-2026 18:16 srv
df-xixt-x 0 0 0 20-Aug-2026 12:38 sys
dfwxixt-xwt 0 0 4096 20-Aug-2026 12:45 tmp
dfwxixt-x 0 0 4096 2-Jul-2026 18:16 usr
dfwxixt-x 0 0 4096 15-Jul-2026 21:56 var
lwxixt-xwt 0 0 7 16-Jun-2026 15:44 bin
-rwxixt-x 0 0 2836528 26-Jun-2026 02:57 init
lwxixt-xwt 0 0 7 16-Jun-2026 15:44 lib
lwxixt-xwt 0 0 9 16-Jun-2026 15:44 lib64
lwxixt-xwt 0 0 8 16-Jun-2026 15:44 sbin
```

- To copy the backup of the boot sector over the boot sector, select **Backup BS**, validate with Enter, use y to confirm and next OK.

More information about repairing your boot sector under [TestDisk Menu Items](#). The following message is displayed:

```
C:\windows\system32\cmd.exe x Ash_1001@LAPTOP-1C14CNS1: x Ash_1001@LAPTOP-1C14CNS1: x + v
TestDisk 7.2, Data Recovery Utility, February 2024
Please select a destination where the marked files will be copied.
Keys: Arrow keys to select another directory
      C when the destination is correct
      Q to quit
Directory /mnt/d/DIGITAL FORENSIC
drwxrwxrwx 1000 1000 4096 20-Aug-2026 12:47 .
drwxrwxrwx 1000 1000 4096 20-Aug-2026 12:47 ..
drwxrwxrwx 1000 1000 4096 20-Aug-2026 13:14 EXP 2
```

The boot sector and its backup are now both OK and identical: the NTFS boot sector has been successfully recovered.

- Press Enter to quit.

```
C:\windows\system32\cmd.exe x Ash_1001@LAPTOP-1C14CNS1: x Ash_1001@LAPTOP-1C14CNS1: x + v
TestDisk 7.2, Data Recovery Utility, February 2024
Please select a destination where the marked files will be copied.
Keys: Arrow keys to select another directory
      C when the destination is correct
      Q to quit
Directory /mnt/d/DIGITAL FORENSIC/EXP 2/recovered
drwxrwxrwx 1000 1000 4096 20-Aug-2026 13:12 .
drwxrwxrwx 1000 1000 4096 20-Aug-2026 13:16 ..
```

```
C:\windows\system32\cmd.exe x Ash_1001@LAPTOP-1C14CNS1: x Ash_1001@LAPTOP-1C14CNS1: x + v
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
* HPFS - NTFS                2048      1048575      1046528 [EVIDENCE]
Directory /
Copy done! 0 ok, 0 failed
>dr-xr-xr-x  0      0      0 20-Aug-2026 13:08 .
dr-xr-xr-x  0      0      0 20-Aug-2026 13:08 ..

Next
Use Right to change directory, 'h' to hide Alternate Data Stream
'q' to quit, ':' to select the current file, 'a' to select all files
'C' to copy the selected files, 'c' to copy the current file
```

- TestDisk displays ***You have to restart your Computer to access your data*** so press Enter a last time and reboot your computer.